

Person	Height	Weight (kg)	r_{women}	r_{men}	Parameter	Women	Men
A	158	52	0.98	0.02	mean (μ)	$\begin{bmatrix} 160 \\ 55 \end{bmatrix}$	$\begin{bmatrix} 180 \\ 78 \end{bmatrix}$
B	162	56	0.95	0.05	mixing coefficient (π)	0.5	0.5
C	166	60	0.998	0.002	covariance (Σ)	$\begin{bmatrix} 9 & 0 \\ 0 & 16 \end{bmatrix}$	$\begin{bmatrix} 16 & 0 \\ 0 & 25 \end{bmatrix}$
D	175	72	0.10	0.90	$1. \text{NCC}(\mu_{\text{women}}, \Sigma_{\text{women}}) = \frac{1}{(2\pi)^{1/2} \Sigma ^{1/2}} \exp\left(-\frac{1}{2} (A - \mu_{\text{women}})^T \Sigma^{-1} (A - \mu_{\text{women}})\right)$ $= \frac{1}{(2\pi \cdot 0.5)^{1/2} \sqrt{144}} \exp\left(-\frac{1}{2} \left(\frac{166-160}{9}\right)^2 + \left(\frac{60-55}{16}\right)^2\right)$ $= \frac{1}{12} \exp\left(-\frac{1}{2} \cdot 0.51^2\right)$ $= 0.064$		
E	180	78	0.05	0.95			
F	185	84	0.02	0.98			

$$|\Sigma| = \begin{vmatrix} 9 & 0 \\ 0 & 16 \end{vmatrix} = 144$$

$$\text{NCC}(\mu_{\text{men}}, \Sigma_{\text{men}}) = \frac{1}{(2\pi \cdot 0.5)^{1/2} \sqrt{400}} \exp\left(-\frac{1}{2} \left(\frac{166-180}{16}\right)^2 + \left(\frac{60-78}{25}\right)^2\right)$$

$$= 0.000135$$

$$r_{\text{women}} = \frac{0.5(0.064)}{(0.5)(0.064) + (0.5)(0.000135)} = 0.998$$

$$r_{\text{men}} = \frac{0.5(0.000135)}{(0.5)(0.064) + (0.5)(0.000135)} = 0.002$$

$$\pi_{\text{women}} = \frac{0.98 + 0.95 + 0.998 + 0.1 + 0.05 + 0.02}{6} = 0.516$$

$$\pi_{\text{men}} = \frac{0.02 + 0.05 + 0.002 + 0.9 + 0.95 + 0.98}{6} = 0.484$$

$$\mu_{\text{women}}^{\text{height}} = \frac{158 + 162 + 166}{3} = \frac{486}{3} = 162$$

$$\mu_{\text{men}}^{\text{height}} = \frac{175 + 180 + 185}{3} = \frac{540}{3} = 180$$

$$\mu_{\text{women}}^{\text{weight}} = \frac{52 + 56 + 60}{3} = \frac{168}{3} = 56$$

$$\mu_{\text{men}}^{\text{weight}} = \frac{72 + 78 + 84}{3} = \frac{234}{3} = 78$$

$$\mu_{\text{women}} = \begin{bmatrix} 162 \\ 56 \end{bmatrix}$$

$$\mu_{\text{men}} = \begin{bmatrix} 180 \\ 78 \end{bmatrix}$$

$$\mu_{women} = \begin{bmatrix} \frac{485}{3} \\ 56 \end{bmatrix} \quad A = \begin{bmatrix} 158 \\ 52 \end{bmatrix} \quad B = \begin{bmatrix} 162 \\ 56 \end{bmatrix} \quad C = \begin{bmatrix} 166 \\ 60 \end{bmatrix}$$

$$\mu_{men} = \begin{bmatrix} 180 \\ 78 \end{bmatrix} \quad D = \begin{bmatrix} 175 \\ 72 \end{bmatrix} \quad E = \begin{bmatrix} 180 \\ 78 \end{bmatrix} \quad F = \begin{bmatrix} 185 \\ 84 \end{bmatrix}$$

$$\begin{aligned} (A - \mu_{women}) (A - \mu_{women})^T &= \begin{bmatrix} 158 - \frac{485}{3} \\ 52 - 56 \end{bmatrix} \begin{bmatrix} 158 - \frac{485}{3} & 52 - 56 \end{bmatrix} = \begin{bmatrix} -17/3 & -4 \end{bmatrix} \begin{bmatrix} -17/3 & -4 \end{bmatrix}^T = \begin{bmatrix} 289/9 & 68/3 \\ 68/3 & 16 \end{bmatrix} \\ (B - \mu_{women}) (B - \mu_{women})^T &= \begin{bmatrix} 162 - \frac{485}{3} \\ 56 - 56 \end{bmatrix} \begin{bmatrix} 162 - \frac{485}{3} & 56 - 56 \end{bmatrix} = \begin{bmatrix} 1/3 & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 0 \end{bmatrix}^T = \begin{bmatrix} 1/9 & 0 \\ 0 & 0 \end{bmatrix} \\ (C - \mu_{women}) (C - \mu_{women})^T &= \begin{bmatrix} 166 - \frac{485}{3} \\ 60 - 56 \end{bmatrix} \begin{bmatrix} 166 - \frac{485}{3} & 60 - 56 \end{bmatrix} = \begin{bmatrix} 13/3 & 4 \end{bmatrix} \begin{bmatrix} 13/3 & 4 \end{bmatrix}^T = \begin{bmatrix} 169/9 & 52/3 \\ 52/3 & 16 \end{bmatrix} \\ (D - \mu_{men}) (D - \mu_{men})^T &= \begin{bmatrix} 175 - 180 \\ 72 - 78 \end{bmatrix} \begin{bmatrix} 175 - 180 & 72 - 78 \end{bmatrix} = \begin{bmatrix} -5 & -6 \end{bmatrix} \begin{bmatrix} -5 & -6 \end{bmatrix}^T = \begin{bmatrix} 25 & 30 \\ 30 & 36 \end{bmatrix} \\ (E - \mu_{men}) (E - \mu_{men})^T &= \begin{bmatrix} 180 - 180 \\ 78 - 78 \end{bmatrix} \begin{bmatrix} 180 - 180 & 78 - 78 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ (F - \mu_{men}) (F - \mu_{men})^T &= \begin{bmatrix} 185 - 180 \\ 84 - 78 \end{bmatrix} \begin{bmatrix} 185 - 180 & 84 - 78 \end{bmatrix} = \begin{bmatrix} 5 & 6 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 25 & 30 \\ 30 & 36 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \Sigma_{women} &= \frac{1}{3} \begin{bmatrix} 289/9 & 68/3 \\ 68/3 & 16 \end{bmatrix} + \begin{bmatrix} 1/9 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 169/9 & 52/3 \\ 52/3 & 16 \end{bmatrix} = \begin{bmatrix} 459/9 & 40 \\ 40 & 32 \end{bmatrix} \\ &= \begin{bmatrix} 17 & 13.333 \\ 13.333 & 10.667 \end{bmatrix} \end{aligned}$$

$$\Sigma_{men} = \frac{1}{3} \begin{bmatrix} 25 & 30 \\ 30 & 36 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 25 & 30 \\ 30 & 36 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 50 & 60 \\ 60 & 72 \end{bmatrix} = \begin{bmatrix} 16.67 & 20 \\ 20 & 24 \end{bmatrix}$$